TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Detailed Practical Study of Batching, Softener, Spreader, Breaker Card and Finisher Card of Jute Spinning

The preparatory line of the jute mill: how hard, lignified reeds are batched with oil emulsion and piled until they soften, then kneaded in the softener, opened into the first sliver by the spreader, and carded twice — on the coarse breaker card and the finer finisher card.

TE-2102 · SPINNING TECHNOLOGY-I SESSIONAL · STUDY 2 · LEVEL 2 TERM 1

1. Object and method of the study

Object: to study the machines that convert baled jute — long, stiff fibre strands (reeds) of ultimate cells cemented by lignin — into a carded sliver ready for the drawing frames: batching and piling, the softener, the spreader, the breaker card and the finisher card.

Method: follow the same routine as Study 1 — trace the material path, identify the working parts, record settings and production points, and sketch the path. One difference runs through the whole jute line: everything is heavier, coarser and slower-working than cotton machinery, because the fibre arrives in long bundles and must be opened without being cut short.

2. Batching and piling

2.1 What to identify and observe

- **The batch** — how bales are selected and grouped by quality (mark/grade) into a batch for one class of yarn; note the cut lengths to which reeds are cut for the machines (the batch is made up to the length the cards can handle).
- **The emulsion plant** — the mixing tank where batching oil is emulsified with water (with an emulsifier) into a stable milky emulsion; record the oil-to-water ratio in use.
- **Application** — the emulsion is sprayed or brushed onto the fibre as it is spread in layers on the floor or passed on a conveyor; find where and how uniformly it is applied.
- **The pile (bin)** — the layered batch is stacked and left to condition, commonly 24–48 hours, so the emulsion penetrates and the fibre softens throughout; note the pile size and the resting time used in

the lab/mill.

> **WORKED CHECK — BATCHING EMULSION QUANTITIES**
>
> Batch of 200 kg jute. Emulsion applied at 30% on the weight of fibre = 0.30 × 200 = **60 kg of emulsion**. With oil : water = 1 : 5, the oil in it = 60 × 1/6 = **10 kg**, i.e. an oil add-on of 10 ÷ 200 = **5% on the weight of fibre**. Check in the lab: change any one figure (batch weight, application %, ratio) and the other two must still agree — this is the calculation the batching department does daily.

3. The softener

3.1 Parts to identify

- **Feed table/lattice** — the piled, oiled reeds are spread by hand into an even sheet and fed forward.
- **Fluted roller pairs** — the long series of pairs of heavy, spring-loaded **fluted (grooved) rollers**; the sheet is carried between them and repeatedly bent and crushed as the flutes of one roller press into the grooves of the other. Count the pairs on the lab machine and record the number.
- **Delivery end** — the softened sheet leaves in a continuous mat; note that the softener forms **no sliver** — it only kneads.

Action to state in the report: the repeated flexing breaks down the stiff, woody bonding of the fibre bundles and works the emulsion through them, so the reeds become pliable enough for the pinned rollers of the spreader and cards to open without breaking the fibre.

3.2 Observations to record

ITEM	WHAT TO WRITE IN THE REPORT
Roller pairs	Number of pairs; whether flutes are straight or spiralled; how the top rollers are loaded
Feed sheet	Evenness of the hand-spread sheet and the approximate feed weight per unit area
Condition of delivery	Softness/flexibility of the mat compared with the unsoftened reed — the direct evidence the machine is doing its work

4. The spreader

4.1 Parts to identify

- **Feed lattice** — carries the softened sheet into the machine.
- **Pinned feed rollers** — pairs of rollers clothed with strong pins that grip the sheet; the first pair feeds slowly, the second runs faster, so the sheet is drawn out and opened between them.
- **Retaining and opening action** — where the pinned surfaces begin to separate the reeds into a loose, continuous web — the **first sliver** of the jute line is formed here.
- **Delivery and coiling** — the sliver is condensed and coiled into a can or rolled onto a drum, ready for the breaker card.

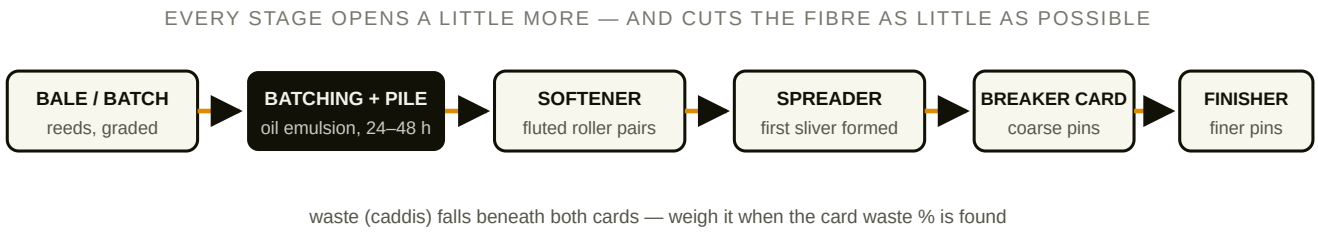


Figure 1 — The jute preparatory line up to the finisher card. Softening comes before opening: dry, unbached jute put straight to the cards would break into short fibre and dust.

5. The breaker card

5.1 Parts to identify

- **Feed lattice and feed rollers** — the pinned feed rollers that present the spreader sliver/sheet to the cylinder; note the strong, coarse pinning.
- **Cylinder** — covered with pinned lags (staves) instead of flexible card clothing; the pins are coarse and angled (raked) to grip the bundles.
- **Workers and strippers** — pairs of smaller pinned rollers working against the cylinder: the worker takes fibre from the cylinder, opens it against the stripper, and the stripper returns it to the cylinder. Count the pairs.
- **Doffer and fancy** — the doffer collects the opened fibre into a web; the fancy (a brush-like roller with long pins/bristles) raises the fibre in the doffer clothing so the web can be stripped and condensed into sliver.
- **Waste fall (caddis)** — the space beneath the machine where short fibre, dust and woody matter drop; find how it is cleared.

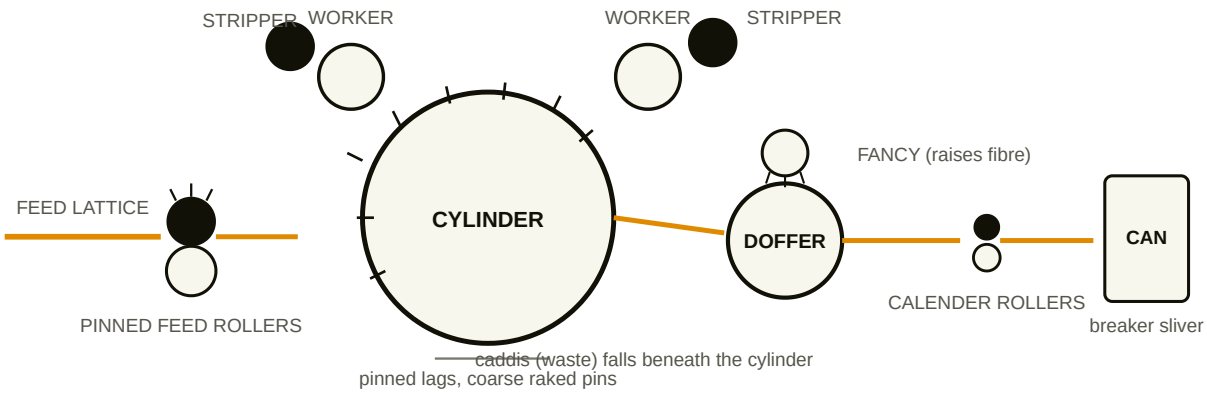


Figure 2 — Cross-section of a jute card (breaker; the finisher is built the same way with finer pinning). Worker–stripper pairs work the fibre repeatedly against the cylinder before the doffer collects it as a web.

5.2 Settings and observations to record

ITEM	WHAT TO WRITE IN THE REPORT
Pinning	Pin density impression (pins per unit area) and pin angle on feed rollers, cylinder and workers — breaker card values
Worker/stripper pairs	Number of pairs; the settings (distances) of worker to cylinder and stripper to worker
Speeds	Cylinder rpm; relative surface speeds of worker (slow) and stripper (fast) compared with the cylinder
Waste	Caddis collected over a timed run as % of feed
Delivered sliver	Sliver weight in the jute form (lb per 100 yd) by weighing a measured length

6. The finisher card

6.1 Parts to identify — and how it differs from the breaker

- **Same skeleton, finer tools** — feed rollers, cylinder, worker–stripper pairs, doffer(s) and fancy are all present, but the pinning is **finer and closer-set** than the breaker's, and there are generally **more worker–stripper pairs**, so the fibre is worked more times and more gently.
- **Feed** — the finisher is fed with breaker-card sliver, often with several ends doubled behind the feed rollers; count the cans feeding the lab machine.
- **Delivery** — many finisher cards carry **two doffers**, delivering two slivers; check whether the lab machine delivers one or two, and record the sliver weight of each.

- **State of the fibre** — compare finisher sliver with breaker sliver by hand: the finisher product is finer-divided, cleaner and more parallel — the opening work of the line is complete, and drafting (drawing frames, TE-2202) takes over from here.

> **WORKED CHECK — FINISHER CARD PRODUCTION**
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> Delivery speed 20 yd/min; delivered sliver weight 7 lb per 100 yd. Length delivered per hour = 20 × 60 = 1,200 yd. Production = 1,200 ÷ 100 × 7 = **84 lb/h ≈ 38 kg/h**. Repeat the calculation with the lab machine's own delivery speed and sliver weight, and compare breaker and finisher outputs — the finisher, fed with already-opened sliver, is normally the faster-delivering of the two.

FEATURE	BREAKER CARD	FINISHER CARD
Feed	Spreader sliver/sheet of cut reeds	Breaker-card sliver (often doubled ends)
Pinning	Coarse, widely spaced, strongly raked	Finer, closer-set
Worker–stripper pairs	Fewer	More — fibre worked more times
Waste (caddis)	Heavy — first removal of short fibre and woody matter	Lighter
Delivered sliver	Coarse, still bundle-like	Finer, cleaner, more parallel

7. What the lab report must contain

SECTION	CONTENT
Batching record	Batch composition, emulsion ratio, application % and piling time, with the emulsion calculation
Material path	Traced path through all five stages with sketches (Figures 1–2 as models)
Parts tables	Each working part and its function in one line, per machine
Settings record	Observed pin densities, pair counts, speeds and settings — from the machines, not from a book
Comparisons	The breaker–finisher comparison table filled from observation, and a jute–cotton contrast (preparation, pinning, waste) with Study 1
Production checks	Emulsion quantities and finisher-card production from your own figures

8. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why does jute need batching oil and a long piling rest, while cotton goes almost directly from bale to blow room?
2. What is the difference in action between the softener and the spreader?
3. State three differences between a breaker card and a finisher card.
4. A batch of 200 kg of jute takes emulsion at 30% on its weight, mixed oil : water = 1 : 5. How much emulsion is made, and what is the oil add-on as % of fibre weight?
5. Why are jute card pins coarse and raked, where cotton card clothing is fine wire?
6. What is caddis, where does it come from, and what does an abnormally high caddis % suggest?

Answer key

1. Jute fibre is not a single soft cell but a long bundle of short ultimate cells cemented by lignin and pectins; dry, it is harsh, woody and brittle, and pinned rollers would shatter it. The oil emulsion lubricates and softens the bundles, and piling gives it time to penetrate evenly, so the fibre becomes pliable enough to card and draft. Cotton is already a soft single cell and needs no softening.
2. The **softener only kneads**: its fluted roller pairs bend and crush the oiled reeds until they are flexible — it forms no sliver. The **spreader opens**: its pinned rollers grip the softened sheet and draw it out, separating the reeds into the first continuous sliver of the line.
3. Any three: breaker pinning is coarser and more widely spaced; breaker has fewer worker–stripper pairs; breaker takes the heavier waste (caddis) from raw spreader feed; breaker delivers a coarser, still bundle-like sliver; finisher pinning is finer, it is fed with breaker sliver (often doubled), and delivers a finer, cleaner sliver.
4. Emulsion = 0.30 × 200 = **60 kg**. Oil = 60 × 1/(1+5) = **10 kg**, which is 10 ÷ 200 = **5% on the weight of fibre**.
5. Jute is fed as long, coarse bundles that must be gripped firmly and torn apart progressively; coarse, strong, raked pins survive and perform that gripping without cutting across the bundles. Cotton fibres are fine single cells, opened by the dense, fine points of wire clothing — jute pins on cotton would destroy it, and cotton clothing on jute would be stripped bare.
6. Caddis is the waste that falls beneath the jute cards — short fibres, dust, bark and woody fragments shaken out during opening. A rising caddis % suggests the fibre is being broken up (too severe an action, too dry a batch, or worn/damaged pins tearing instead of opening), because sound long fibre should pass forward into the sliver.